

MORRIS SARDO

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<https://morris-sardo.github.io/>

PROFILE

MSci Computer Science (Artificial Intelligence) student at Royal Holloway, University of London, with a strong interest in machine learning research grounded in mathematical modelling, optimization, and model reliability. My academic training has progressively shifted from software-focused development toward a research-oriented understanding of AI systems, with particular emphasis on generalization, evaluation, and dimensionality reduction.

EDUCATION

Royal Holloway, University of London — Egham, UK

MSci Computer Science (Artificial Intelligence)

September 2022 – June 2026

Degree classification: First Class (1st)

Awarded: the Best Final Year Project Prize 2026

City and Islington College — London, UK

Level 3 Access to Higher Education Diploma (Engineering)

September 2021 – June 2022

Final grade: 139 / 145

City and Islington College — London, UK

GCSE Mathematics

September 2020 – June 2021

Grade: 7

TECHNICAL SKILLS

Programming-Tools

- Languages: Python, Java, C, SQL, JavaScript, HTML, CSS
- Frameworks & Libraries: scikit-learn, TensorFlow, Keras, PyTorch
- Software Practices: Test-Driven Development (TDD), Version Control Systems (GitLab), MVC design pattern
- Documentation & Testing: JUnit, JavaDoc
- Certification: MATLAB Onramp

Machine Learning-Data Science

- Supervised learning: k-Nearest Neighbours, decision trees, linear regression, logistic regression
- Unsupervised learning: clustering methods
- Neural networks for classification and regression
- Convolutional Neural Networks (CNNs) and fully connected architectures
- Overfitting and underfitting analysis and mitigation
- Model optimisation through cost-function minimisation
- Performance evaluation: cross-validation, validation sets, out-of-bag estimation
- Statistical testing: p-values, null hypothesis testing
- Dimensionality reduction: PCA, Multidimensional Scaling (MDS), t-SNE

Advanced Topics

- Recurrent Neural Networks (RNNs) and LSTM architectures
- Transfer Learning
- Generative Adversarial Networks (GANs)
- Reinforcement Learning
- Natural Language Processing
- Online Machine Learning (extracurricular study)

RESEARCH EXPERIENCE

Research Placement – Royal Holloway, University of London

June 2024 – September 2024

- Collaborated with a research group investigating the visualisation of high-dimensional datasets mapped to two-dimensional representations while preserving data reliability.
- Studied and applied linear and non-linear dimensionality reduction techniques, including PCA, t-SNE, ISOMAP.
- Implemented dimensionality reduction algorithms in Python and evaluated their reliability and stability.
- Applied Support Vector Classification (SVC) to reduced representations.
- Assessed algorithmic performance and robustness under different parameterisations.
- Participated in weekly research meetings to discuss methodology, progress, and next research steps.

TECHNICAL SKILLS

Medical Image Segmentation with Attention-Enhanced U-Net Architectures

Final Year MSci Project

- Research-focused project addressing medical image analysis through deep learning-based image segmentation.
- Implemented a U-Net backbone augmented with external attention gate mechanisms to enhance feature selection and spatial focus.
- Designed and experimented with multiple configurations and combinations of attention gates to study their impact on segmentation performance.
- Conducted systematic experiments to evaluate how attention mechanisms improve image readability, structural clarity, and interpretability in medical imaging contexts.
- Analysed model behaviour and performance across architectures to identify configurations that improve segmentation quality.
- Project emphasises robustness, interpretability, and applicability of deep learning models in medical imaging.

PROJECTS

- Machine Learning Classification Projects: Implemented and evaluated supervised classification models using conformal and non-conformal prediction methods, focusing on model efficiency and generalisation.
- Deep Learning Project: Designed and trained a CNN and fully connected neural network for classification tasks, analysing overfitting behaviour and improving generalisation performance.
- Financial Modelling: Implemented Black–Scholes, Binomial Tree, and Monte Carlo methods to price European and American call and put options.
- Optimisation Methods: Applied integer programming, branch-and-bound, LP duality, simplex methods, and combinatorial optimisation techniques to structured optimisation problems.

RESEARCH INTERESTS

- Machine learning reliability and generalisation
- Dimensionality reduction and representation learning
- Trustworthy and interpretable AI
- AI applications in health, media, and inclusive technologies

References: Available on request.